

Victor Delaney Sire

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EDUCATION

University of Michigan, Ann Arbor, MI

Expected Graduation: May 2027

Bachelors of Science in Engineering, Mechanical Engineering

Relevant Coursework: Solid Mechanics, Dynamics and Vibrations, Design and Manufacturing, Mechanical Behavior of Materials, Fluid Mechanics, Thermodynamics, Introduction to Aerospace Engineering

Thermodynamics in Paris (University of New Haven)

May 2025 - July 2025

- Studied thermodynamics for two months in Paris, France. I attended lectures of Dr. Frederick De Meyer, completing assignments, exams and projects. Researched and wrote a paper on nuclear fission in power plants and GenIV reactors
- Engaged in a cultural immersion focused program visiting engineering-related museums and sights such as Le Musee des Arts et Metiers and architecturally innovative structures offering a greater historical perspective of engineering

PROJECT EXPERIENCE

Motorized Linkage

December 2025 - May 2026

- Designed a motorized linkage mechanism to automatically detect and illuminate moving targets, using CAD and multibody simulation (MBS) to achieve $\geq 60^\circ$ angular displacement, $< 30^\circ$ transmission-angle deviation, and a 1.5 s displacement time
- Optimized linkage dynamics using inertia matching, balancing transmission performance, $> 80\%$ mechanical-efficiency target, and ≤ 10 lb total system weight
- Manufactured and assembled the mechanism through mill/lathe machining, 3D printing, and waterjet cutting; integrated motor, limit switch, photosensor, and Arduino circuitry for automated operation
- Developed PID-based motion control and iterated controller parameters through testing to improve accuracy, stability, and response time; analyzed hard-stop loading and > 240 MPa link stresses

Robotic Mobility Platform (RMP)

January 2025 - April 2025

- Designed and modeled an RMP using SolidWorks, optimizing mechanical components within 10in.x10in.x12in. dimensional constraints such as rack and pinion and magnetic extension for object retrieval and placement in dynamic environments
- Led end-to-end fabrication of the RMP using manual mill and lathe machining, waterjet cutting, and 3D printing, used SolidWorks software to create engineering drawings ensuring precision tolerances and functional integration of all mechanical systems while gaining experience calculating error propagation and tolerance stackup
- Competed with RMP in a robotics competition using a control box, where the RMP was tasked with collecting a flag, engaging with several buttons and switches, and finally depositing the flag, placing top 10% in the competition

UMichigan Solar Car

August 2023 - May 2024

- Researched and co-designed a trochoidal steering system for the 3000km 2023 Bridgestone World Solar Challenge in Australia, contributing to a 4th-place global finish with smoother and more reliable handling/maneuverability
- Engineered a ballast box to comply with American Solar Challenge regulations, ensuring structural integrity and safety through iterative design and regulatory alignment without impairing comfort or performance of the solar vehicle
- Utilized Siemens NX for CAD, Altair Adams for dynamic simulation, and HyperWorks for FEA, using these advanced tools to validate and refine solar vehicle steering arms

Laser-cut Hovercraft

December 2023 - May 2024

- Designed and built a functional light-weight hovercraft using CATIA for CAD modeling and STAR-CCM+ for aerodynamic simulations, optimizing lift and stability through iterative design analysis and controls via arduino programming
- Fabricated key components including a 3D-printed impeller and laser-cut frame, translating digital models into a working prototype and broadening my understanding of Gcode manufacturing
- Authored a comprehensive engineering report and delivering a formal design review presentation to faculty and peers

WORK EXPERIENCE

UROP Wind Turbine Research (RWTH Aachen University)

Aachen, Germany

Research Intern

May 2026 - July 2026

- Used Design of Experiment (DoE) to set up research objectives, response variables, parameters and experimentation methods to create a sensitivity analysis for energy induced modes of failure in wind turbine roller bearings
- Used Simulink software to simulate wind turbine bearing dynamics using parameters obtained via Latin Hypercube Sampling. Wrote MATLAB and C++ scripts to process simulation data, calculate response variables and create visuals such as Partial Rank Correlation Coefficient heatmaps
- Wrote a short research report and presented a poster detailing my findings at the UROP symposium

SKILLS

Computer Programming: C++, Python, MATLAB, Excel

Engineering Software: SOLIDWORKS, Onshape, Siemens NX, CATIA, STAR-CCM+, Simulink, Ansys, Altair Hyperworks

Manufacturing: 3-D Printing, soldering, drill press, vertical/horizontal band saw, lathe, mill, waterjet

Languages: French (professionally proficient) and Spanish (conversationally proficient)